1. Activation Functions in The Output Layers and Corresponding Loss Functions:

a choice of the activation functions in the output layer and the corresponding loss function is strictly decided by the specific type of problem the neural network is applied to.

2. Binary Classification:

A. a binary classification problem is that its result is probability rather than a continuous numeric value, in binary classification, its result is restricted to a range from 0 to 1.

B. it typically uses Sigmoid function as the activation function in the output layer and Cross Entropy function as the corresponding loss function.

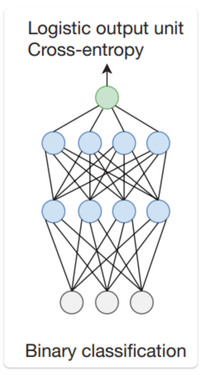
C. logistic sigmoid function: $S(z) = \frac{1}{1+e^{-z}} = \frac{e^{z}}{e^{z}+1}$

D. cross entropy function: $e(\hat{y}_i) = -\frac{1}{m}\sum_{i=0}^{m-1}\left(y_i \cdot \ln(\hat{y}_i) + (1-y_i)\cdot \ln(1-\hat{y}_i)\right)$,

- y_i = desire output y (ground truth) from i-th pair training example which corresponding to single output perceptron
- $\hat{y}_i$ = predicted output P_y from single output perceptron which corresponding to i-th pair training example
- m = number of pair training examples

E. the reason why binary classification utilizes the sigmoid function as the activation function is that the sigmoid function restricts its output to a range from 0 to 1, which allows its output to be interpreted intuitively as probability.

F.



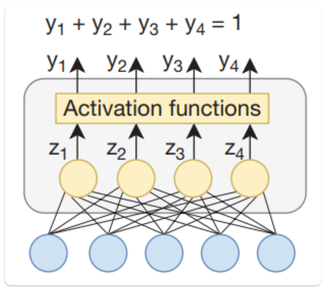
3. Multiclass Classification:

A. a multiclass classification problem is that its results are probability distribution rather than continuous numeric values, in multiclass classification, not only its results are restricted to a range from 0 to 1 but also the sum of its outputs is exactly 1.

B. it typically uses Softmax function as the activation functions in the output layer and Categorical Cross Entropy function as the corresponding loss functions.

C. softmax function: $Soft(z_k) = \frac{e^{z_k}}{\sum_{j=0}^{n-1} e^{z_j}}$

- z_k = the weighted sum function of k-th output perceptron (≡ k-th class) in the neural network
- j = number of output perceptrons (≡ classes) in the neural network

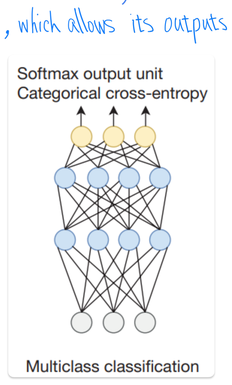


D. categorial cross entropy function: $e(\hat{y}_{ij}) = -\frac{1}{m}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1} y_{ij} \cdot \ln(\hat{y}_{ij})$,

- y_{ij} = desire output y (ground truth) from i-th pair training example which corresponding to j-th output perceptron (≡ j-th class)
- $\hat{y}_{ij}$ = predicted output P_y from j-th output perceptron (≡ j-th class) which corresponding to i-th pair training example.
- m = number of pair training examples
- n = number of output perceptrons (≡ classes) in the neural network.

E. the reason why multiclass classification utilizes the softmax functions as the activation functions is that the softmax functions restrict their outputs to a range from 0 to 1 and guarantees that the sum of its outputs is exactly 1, , which allows its outputs to be interpreted intuitively as probability distribution across multiple mutually exclusive classes.

F.



4. Regression:

A. a regression problem is that its result(s) is/are continuous numerical value(s) (such as dollar amounts) rather than probability, in regression, its result(s) is/are not restricted to a range from 0 to 1 as classification problems.

B. it typically uses Linear function as the activation function(s) in the output layer and Mean Square Error function (MSE function) as the corresponding loss function(s).

C. linear function: it acts as if there is no activation function at all, formally, the activation function is $f(z)=z$, meaning its output is simply raw value of the weighted sum function.

D. mean square error function : $e(\hat{y}) = \frac{1}{m}\sum_{i=0}^{m-1}(y_i - \hat{y}_i)$,

- y_i = desire output y (ground truth) from i-th pair training example which corresponding to single output perceptron
- $\hat{y}_i$ = predicted output P_y from single output perceptron which corresponding to i-th pair training example
- m = number of pair training examples

or

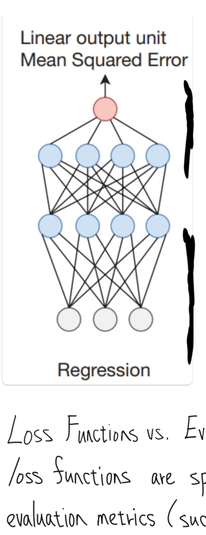
$= \frac{1}{m}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}(y_{ij} - \hat{y}_{ij})^2$,

- y_{ij} = desire output y (ground truth) from i-th pair training example which corresponding to j-th output perceptron (≡ j-th class)
- $\hat{y}_{ij}$ = predicted output P_y from j-th output perceptron (≡ j-th class) which corresponding to i-th pair training example.
- m = number of pair training examples
- n = number of output perceptrons (≡ classes) in the neural network.

E. the reason why regression utilizes the mean square error functions as the activation functions is that the linear functions don't experience the saturated neurons problem, there is no need to use the (categorical) cross entropy functions to avoid the problem. the mean square error function is shown to be a good and effective loss function for the linear function.

F. the reason why regression utilizes the linear functions as the activation functions is that the linear functions don't restrict its output / their outputs to a range from 0 to 1, which allows its outputs to be interpreted intuitively as continuous numerical values (such as dollar amounts)

G.



5. Loss Functions vs. Evaluation Metrics:

A. loss functions are specifically used by learning algorithms to optimize neural networks during the training stage.

B. evaluation metrics (such as "accuracy" for classification or "mean absolute error" for regression) are intuitive indicators for users to evaluate how well neural networks actually performance.